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Prediction of the fibre content of sugarcane stalk by direct scanning using visible-shortwave near infrared spectroscopy

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ABSTRACT

Knowledge of the fibre content in sugarcane stalks is important for breeding programmes and non-destructive measurements for its applications. This study aimed to use portable visible-shortwave near infrared (Vis/SWNIR) with a wavelength range of 570–1031 nm to evaluate the fibre contents of cane stalks. Fibre models were established by partial least squares (PLS) regression using spectra obtained from interactance mode with an absorbance unit of log (1/R). The models were constructed using a sample set of three sample sections (i.e., bottom, middle, and top of the sugarcane stalk). The samples were scanned with different integration times (200, 300, and 400 ms), and the spectra were pre-treated using different pre-processing techniques. The fibre content models were established based on both combined sample sections (CSS model) and individual sample sections (ISS model) obtained from the combination of three sample sections and an individual sample section, respectively. The results showed that the models that were developed using raw spectra with integration time of 300 ms had the best performance. This model had coefficients of determination of the prediction set (r^2) of 0.75, 0.81, 0.81 and 0.71 and root mean square errors of prediction (RMSEP) of 0.81, 0.63, 0.80 and 0.73%fibre for the CSS model and bottom section, middle section, and top section of the ISS model, respectively. These results indicated that the models could be used for screening. Moreover, it was observed that the bottom section model had the lowest RMSEP. The model can be used as a rapid protocol for predicting the fibre content of sugarcane stalks, making it a useful method for a breeder to screen the fibre content in the field when monitoring during breeding programmes.

1. Introduction

Sugarcane is not only the main source in sugar production, but is also an important crop for energy production, as well as for by-products such as ethanol and fibres across the world. Fibre content refers to the solid fibrous part (bagasse) of sugarcane [1]. It mostly contains cellulose, hemicellulose, and lignin (also referred to as the lignocellulosic biomass), which exhibits great potential as a major feedstock for biofuel production worldwide. Consequently, numerous studies have attempted to generate improved sugarcane varieties with the characteristic of a higher fibre content to provide a greater production of biomass [2,3]. Biomass is utilised as fuel to provide steam and electricity to run the sugar mills. Moreover, fluctuating sugar and energy prices have forced sugar mills to implement changes to increase the capacity of power plants [4], because most sugar mills currently also sell electric power [5].

Knowledge of fibre content is important for breeding programmes because it is used to develop sugarcane varieties for specific uses and that are adapted to different climatic conditions. For example, these include plants with high fibre content and cell walls that are easily broken, favouring the production of ethanol from bagasse [6]; plants with a small stalk diameter and high fibre content, which increase the strength of sugarcane for use in windy areas; and others.

To increase the potential for obtaining varieties that provide the optimum fibre content for specific uses, breeders must monitor fibre content during sugarcane growth and study its response to changes in the environment. However, the traditional measurement of fibre content destroys the sugarcane stalks. Because of this, breeders require a large experimental area that is sufficient to produce enough stalks that can be cut monthly for experiments. However, there are differences in

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Nomenclature

Vis/SWNIR	Visible-shortwave near infrared
NIR	Near infrared
CSS	Combined sample sections
ISS	Individual sample sections
PLS	Partial least squares
PCA	Principal component analysis
R^2_{max}	Maximum coefficient of determination
R^2	Coefficient of determination of calibration set
r^2	Coefficient of determination of prediction set
RMSEC	Root mean square error of calibration
RMSEP	Root mean square error of prediction
Bias	Average of difference between the measured values and the predicted values
RPD	Ratio of the standard deviation of the reference data to root mean square error of prediction
rep	Repeatability
SD_y	Standard deviation of the reference data
SNV	Standard normal variate
MAS	Moving average smoothing
D^1	First derivative
D^2	Second derivative

each sugarcane sample, even if they are grown from the same seed variety and planted in a controlled field. These weaknesses have led to a low percentage of success in such breeding programmes. Furthermore, to investigate the prevention of lodging that can increase sugar yield by 15–35% [7], breeders have to measure the fibre content at the bottom, middle and top sections of each stalk, since the characteristic of fibre content accumulation has a different pattern for three sub-sections [8]. The fibre content reflected the strength of the stalk against lodging, and its accumulation pattern depended on the sugarcane cultivars. All of these constraints are costly and time-consuming.

Recently, techniques have been developed that can measure the agricultural material without destruction, such as spectroscopic methods [9–15]. This technology has been employed for the non-destructive measurement of fruit quality [16–18] and Pol in cane [19]. In addition, it has been used to follow up on the quality of fruit, such as for determining the ripe-stage eating quality of mangos [20,21]. Thus, this technique is an interesting method determining fibre content.

In 2014, a similar study reported the application of a spectroscopic method to predict the fibre content of sugarcane stalks that could rapidly obtain the fibre content [22]. However, this application had to be conducted inside a light-proof measurement box. As such, it cannot be used as a non-destructive method for breeding programmes in the field.

Consequently, this study focused on the non-destructive measurement of fibre contents of sugarcane stalk using a spectroscopic method that would provide tangible data for following up in breeding programmes. Notably, if the breeders can measure the fibre content rapidly in the same stalks until the end of experiments, it is easier for studying the effects on fibre content over a short time. It also takes less time and reduces the trial size in experiments, leading to the possibility of a breeding programme based on a powerful control system.

Because of the non-homogeneity of sugarcane texture, differences in peel, and the variety of skin colours, integration times and pre-processing techniques were needed for this investigation [23]. As such, the primary aim of this study was to investigate the ability of a portable Vis/SWNIR instrument to predict the fibre content of sugarcane stalks. In addition, specific study objectives were as follows: (1) to compare the performances of different integration times including short-time (200 ms), mid-time (300 ms), and long-time (400 ms) using partial least squares (PLS) analysis; (2) to compare the performances of different pre-processing techniques on PLS analysis; and (3) to compare the performance of the combined sample sections model (CSS model) and individual sample sections model (ISS model).

2. Materials and methods

2.1. Sample preparation

A total of 62 sugarcane stalks (after 8–12 months of growth) were supplied from the Khon Kaen Field Crops Research Centre (KKFCRC) during the 2016/2017 harvest season. They consisted of 6 common varieties (KK3, LK92-11, TP1, TP2, UT-15, and K07-037) that are grown in north-eastern Thailand (a major plantation area). To ensure that the

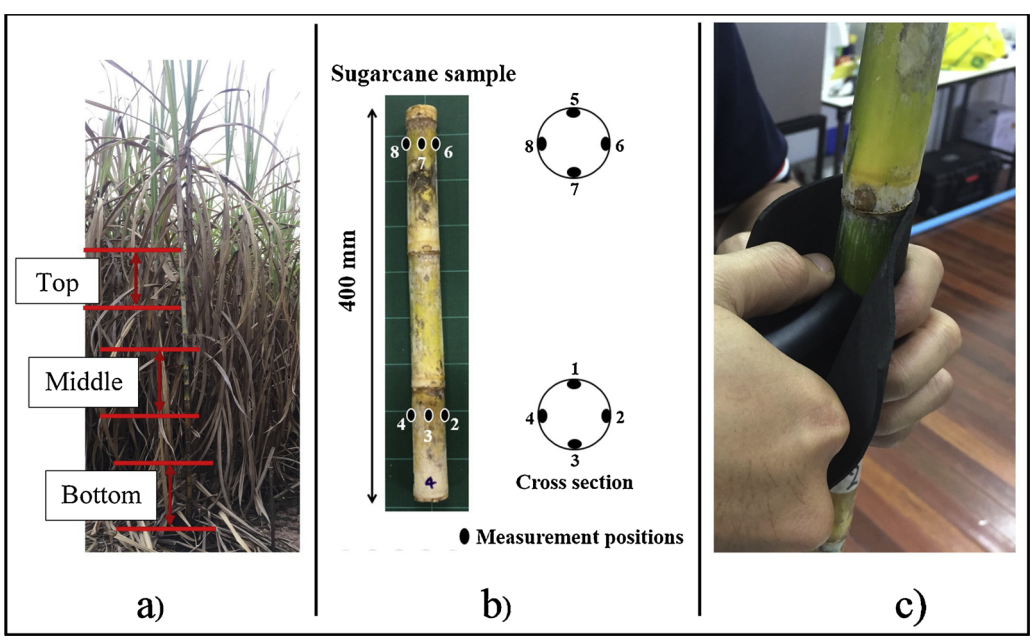

Fig. 1. a) Top, middle and bottom sections of cane stalk, b) measurement positions of Vis/SWNIR spectra on sugarcane samples, c) sample scanning.

model was developed using a wide range of fibre contents to obtain a robust model, several varieties of samples were used, as well as samples at different stages of maturity. Each cane stalk was cut 10 cm above the ground and was cut at the top of stalks (near the growing point). Afterward, all leaves were removed, and the sugarcane stalk was divided into three sub-sections (i.e., top, middle, and bottom) with lengths of approximately 400 mm (see in Fig. 1a). Thus, the total number of samples equals to 186 (62 stalks × 3 sub-sections). Finally, the samples were taken to the laboratory and kept at room temperature (25 ± 2 °C) for an hour before the experiment. Moreover, to examine the precision of the fibre content values of the reference data, 10 samples were collected from the same plantation area and harvest season to measure the fibre content two times using the reference method, thus creating a duplicate for each sample. In this research, the length of 400 mm was designed to obtain a fibre content value that represented each sub-section, whereas the method of separating the stalk into three sub-sections was applied to achieve the same stalk positions in the sample as in the breeding programmes.

2.2. Spectral acquisitions

The cane surface of each sample was directly scanned (see in Fig. 1c). The scan was performed using a portable visible-shortwave near infrared (Vis/SWNIR) spectrometer (HNK Engineering, Hokkaido, Japan) with an interactance mode across a wavelength of 570–1031 nm at 1 nm intervals. This was done inside the laboratory at a room temperature of 25 ± 2 °C. A 25 W halogen lamp was used as a light source, and fibre optics were also used for the signal transmission. All spectral data were stored in a computer.

In this experiment, relative absorbance compared to the standard plate was used instead of absolute absorbance. The relative absorbance was calculated from the reflection of intensity to the Teflon plate. It was designed to optimize the measurement condition with different integration times, since the light detector has the sensitivity of the dynamic range. Prior to scanning the samples, a standard black sheet (Ethylene-vinyl acetate, EVA) and a 2 mm-thick Teflon plate were scanned as reference materials, as shown in Fig. 2. The absorbance was calculated using the following formula:

$$R = \frac{\text{Reflectance}_{\text{sample}} - \text{Reflectance}_{\text{black sheet}}}{\text{Reflectance}_{\text{teflon plate}} - \text{Reflectance}_{\text{black sheet}}} \quad (1)$$

$$\text{Absorbance} = \text{Log}(1/R) \quad (2)$$

Each sample was scanned at eight positions using the integration times of 200, 300, and 400 ms. During the measurements, the gap between the scanned point and the light window was covered with a black sheet (Ethylene-vinyl acetate, EVA) to shield the area from external light.

Fig. 1b illustrates the measurement position. The first position was chosen at random on the upper part of the sample below the growing point at approximately 20 mm. The next three measurement positions were performed 90°, 180°, and 270°, respectively, moving around the sample from the first position [19]. Next, four spectra (points 5 to 8), were acquired at the lower part in the same manner as the upper part. Then, the spectrum of each sample was computed by averaging the eight successive scans.

2.3. Reference analysis

The fibre content was determined using Method 4 [24]. After the scanning was finished, each whole sample was ground to become a fibrous cane. Then, the samples were washed in a cotton cambric bag for three 6-minute cycles in an automatic washing machine. Finally, the samples were dried at 105 °C until they maintained a constant weight. The fibre content (%fibre) was determined using the following formula:

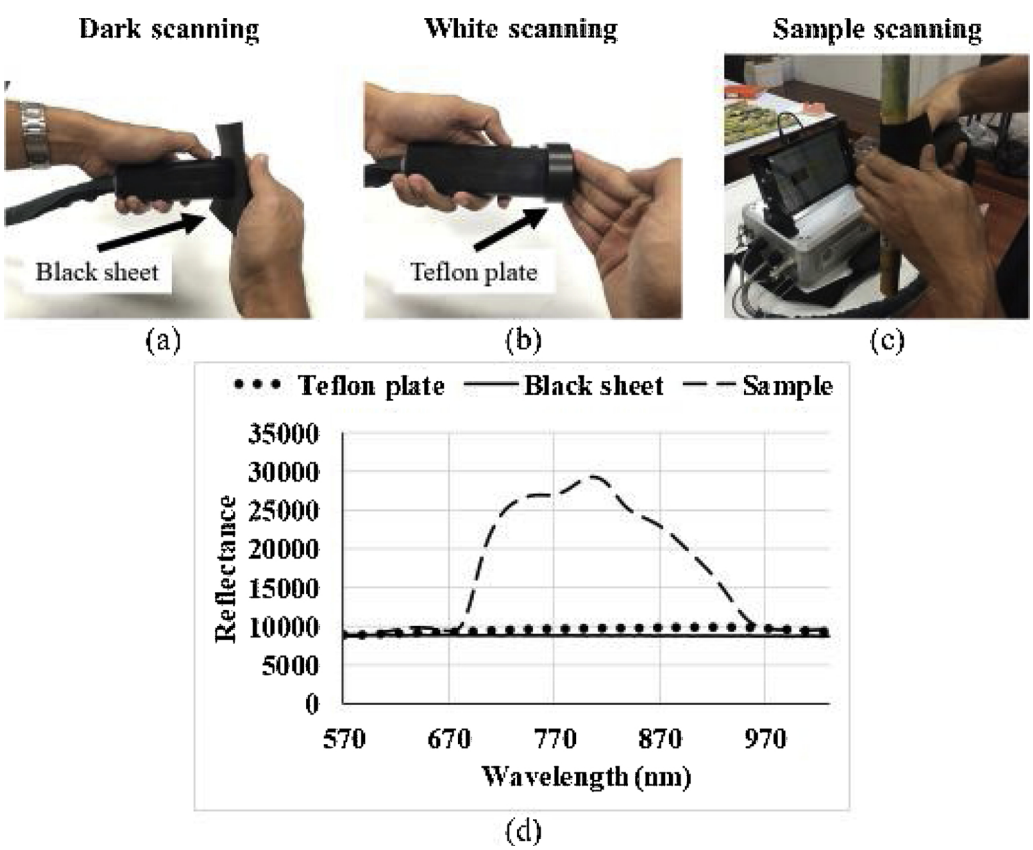

Fig. 2. Spectral acquisitions of cane stalk (a) dark scanning, (b) white scanning, (c) sample scanning, and (d) reflectance graph.

$$Fibre\ content\ (\%fibre) = \frac{W_f}{W_i} \times 100 \quad (3)$$

where W_i is the initial weight of cane sample and W_f is the final weight after drying in the oven.

2.4. Repeatability and maximum coefficient of determination

The precision of the reference method is indicated by the repeatability (rep) value. Prior to determining the fibre content, each whole sample was ground to become a fibrous cane. After that, the ground sample was mixed and was randomly spilt into two halves in order to get the homogeneous samples. Finally, determination of the fibre content was done for the duplicates and the repeatability of fibre content was calculated as the standard deviation of the difference between the duplications, and the maximum coefficient of determination (R^2_{max}). The R^2_{max} was calculated using the following formula:

$$R2max = \frac{SD_y^2 - rep^2}{SD_y^2} \quad (4)$$

where SD_y is the standard deviation of the reference data. The R^2_{max} is the maximum of the coefficient of determination (R^2).

The R^2_{max} value stated that if there is no error in the cane stalk spectra, the R^2 could be equal to R^2_{max} [25]. R^2_{max} will be low if the difference between the duplicates is high. However, inhomogeneity causes significant sampling error, and sampling error causes low R^2_{max} and R^2 values. Meanwhile, r^2 (coefficient of determination of the test set) depends on the validated spectra. More error leads to low r^2, and therefore, inhomogeneity of the cane sample strongly affects the prediction.

2.5. The spectral data pre-processing

To establish the models, the spectral data that were used for model development were either pre-processed or not. The following pre-processing techniques were used for modelling: baseline offset, mean centering, standard normal variate (SNV), range normalization, moving average smoothing (MAS) (segment size: 5 nm), MAS (segment size: 5 nm) + first derivative (D^1) (segment size: 21 nm), and MAS (segment size: 5 nm) + second derivative (D^2) (segment size: 21 nm). In this study, the pre-processing techniques were performed using the MATLAB program (MathWorks version 2017, Massachusetts, USA). These pre-processing techniques were used to reduce the effects of light scattering and to eliminate changes the light-path lengths of the samples, which can lead to the appearance of baseline shifts and multiplicative scattering effects in the spectra.

2.6. Data analysis

Before developing the calibration model, principal component analysis (PCA) was applied to detect any spectral outliers that may affect the performance of the model [26]. In this research, CSS and ISS models were developed using partial least square (PLS) regression, which was performed using in-house code operated in the MATLAB program.

The optimal number of PLS factors was determined by a selection of the minimum on RMSEP [27]. In addition, the trend of RMSEC was considered to avoid over-fitting by choosing the number of PLS factors at the point of value of the RMSEC, which was gradually unchanged and provided the smallest gap in differences between the RMSEC and RMSEP.

To evaluate the model performance, a total of 62 stalks was sorted by the average fibre content of each stalk, which were then separated into a calibration set and a prediction set. The prediction set was selected by taking one of every four stalks, except those with the lowest and highest fibre contents [28]. Then, 75% of the stalks (47 stalks) were designated as the calibration set, whereas 25% (15 stalks) were designated as the prediction set. Accordingly, the calibration set included 141 samples (47 stalks × 3 sub-sections) and the prediction set included 45 samples (15 stalks × 3 sub-sections). This selection aimed to ensure that each set could cover the entire range of fibre contents and obtain similar statistical characteristics.

The performance of the models was evaluated using the following indices: the coefficient of determination of the calibration set (R^2), the coefficient of determination of the prediction set (r^2), the root mean square error of calibration (RMSEC), the root mean square error of prediction (RMSEP), bias, and the ratio of the standard deviation of the reference data to root mean square error of prediction (RPD). A good model should exhibit the highest r^2 and the lowest RMSEP. Zornoza et al. [29] recommended that the model should provide the following: (1) an excellent prediction can be made when $R^2 > 0.90$ and when RPD > 3; (2) a good prediction can be made when $0.81 < R^2 < 0.90$ and 2.5 < RPD < 3; (3) only an approximate prediction is permitted when $0.66 < R^2 < 0.80$, and 2.0 < RPD < 2.5; and (4) only a poor prediction can be made when $R^2 < 0.66$ and RPD < 2.

For CSS model development, combinations of the three sub-sections (i.e., top 47, middle 47, and bottom 47 samples) were used, resulting in 141 samples for calibration. This method used spectra obtained from three integration times across eight pre-processing techniques for modelling. To compare the spectral preprocessing ability (factor 1) and integration time (factor 2) in prediction, ANOVA (analysis of variance) was performed. Considering that each condition was set without replication, they should then be analysed using two-factor without replication analysis, with significant difference at 5% level, to test the influence of the two-factor on different RMSEP values. This was performed across a randomized complete block design (RCBD).

Eight spectral pretreatments were tested for whether they produce significantly different results. Meanwhile, integration times might differ slightly in their RMSEP. Thus, integration times could be a nuisance. So, they were assigned as a blocking factor [30]. Three levels of integration time (200, 300, and 400 ms) were large enough to try out for the influence of eight spectral pretreatment methods. An RCBD could be used and each integration time was treated as a block. Hence, three blocks were used (integration times) and spectral pretreatments were randomly determined. The null hypothesis (H0) was that RMSEP is not significantly different among eight different spectral preprocessing methods, at a 95% confidence level. If this resulted in the null hypothesis being rejected, the difference in RMSEP of each spectral pretreatment would be assigned to Duncan's test. The models were evaluated using 45 samples of the combined prediction sample set.

For ISS model development, 47 samples for each of the three sections were separately used to run the model as the ISS model. The models were evaluated using 15 samples of each prediction sample set.

3. Results and discussion

3.1. An overview of the spectra and statistical characteristics of the fibre content

The spectra of short-wave bands have more the penetration depth than the long-wave bands, allowing a more adequate sampling of inhomogeneous material [31]. Qin and Lu [32] found that the light penetration depth in fruit tissue depended on the type of fruit, ranging from 7.1 to 62.5 mm. Hence, the Vis/SWNIR bands applied for the sugarcane tissue might satisfy, even though it is an inhomogeneous material and its surface is thick and hard. To ensure that the spectra would represent the absorbance of the whole sample, each sample was scanned at eight positions. The raw spectra of sugarcane stalks obtained from the integration time of 300 ms are shown in Fig. 3a. Several different trends were observed in the absorbance of visible region spectra at 571–650 nm because the various skin colours of the sugarcane stalks impacted the sensitivity of the visible regions, which is similar to results

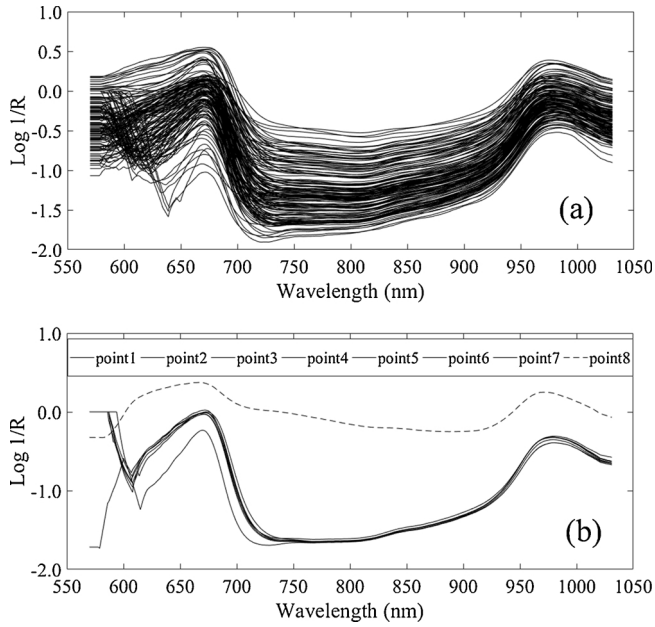

Fig. 3. Raw Vis/SWNIR spectra of cane stalks (a) total samples (b) sample numbers 53.

discussed by Jamshidi et al. [27] and Lui et al. [33]. Thus, this spectral region was considered to be noise and were excluded from the modelling. Obvious peaks were observed at approximately 980 nm and were related to the second overtone of the O—H stretching mode [34]. In this research, it was found that the reflectance of the sample scanning was higher than the reflectance of the Teflon plate scanning (see in Fig. 2), which made the absorbance (log1/R) of some wavelengths display negative values. This finding may be a result of the high crystallinity of fibres in sugarcane (with a crystallinity index in the range 63–68% [1]), which may cause abundant light scattering in the sample. This phenomenon leads to the result of the longer integration times displayed in lower absorbance as shown in Fig. 4a. According to the various skin colours of cane stalks, orange and red skin colours of some samples affected the absorption of spectra at 620–650 nm [35]. Fig. 4b shows the spectra of the three sub-sections obtained from the integration time of 300 ms. The bottom section obtained the highest absorbance, which is because it contained the highest percentage of sugarcane juice [36]. The top sections on the other hand had the lowest absorbance. The spectra of the three sub-sections were different, which might lead to the conclusion that each sub-section had unique spectra of its own. These might affect the result of the CSS model. Before modelling, the spectra were checked by PCA. Two outliers were detected and then removed. The dashed line in Fig. 3b. is the spectrum detected as the outlier. This spectrum is different from the other spectra obtained from the same sample. They came from bent or wounded stalk. As such, the users should be careful when performing scanning. For the modelling, it was determined that spectra in the range of 700–1000 nm with intervals of 3 nm were optimal. These findings were selected using the principles of selection of effective wavelengths recommended by Liu et al. [33], who used Vis/SWNIR spectroscopy to determine the contents of soluble solids and the pH levels of rice vinegars.

Table 1
Statistical characteristics of the samples for the calibration and prediction sets.

	Fibre content (% Fibre)	CSS model	ISS model		
		All sample sections	Bottom	Middle	Top
Calibration set	N	141	47	47	47
	Max	15.30	14.20	15.30	14.04
	Mean	11.26	11.41	11.07	11.30
	Min	7.10	7.30	7.10	8.20
	SD	1.65	1.64	1.87	1.43
Prediction set	N	45	15	15	15
	Max	15.07	14.02	15.07	13.90
	Mean	11.35	11.47	11.23	11.34
	Min	8.09	9.00	8.09	8.90
	SD	1.62	1.53	1.92	1.48

N: number of samples, max: maximum, min: minimum, mean: average, SD: standard deviation.

The repeatability (rep) and the R^2_{max} of reference method were 0.30%fibre and 0.95, respectively. These results indicated that the precision of the reference method was good. A summary of the statistical characteristics of the samples for the calibration and prediction of the data sets. The number of samples (N), maximum (max), average (mean), minimum (min), and standard deviations (SD) of the fibre content determined by the conventional method, as shown in Table 1. The fibre content for all samples used for developing the CSS model had a wide range of between 7.10–15.30 %fibre because samples from six varieties and samples at several different stages of maturity were used. Moreover, in both calibration and prediction sets, the range, average, and SD of the fibre contents were almost identical.

The samples applied to establish the ISS model that separated all the sample sections into the bottom, middle and top sections had differences on some statistical characteristics of fibre content. The average value was similar, but they had different SD values. The middle section had the highest SD value, while the top section had the lowest, which led to difference of R^2 and r^2.

3.2. PLS models established based on combined sample sections (CSS models)

The CSS models were accomplished using 141 samples, including three sub-sections, where each sample contained the average of eight spectra. An effective model was determined by generating the PLS model with different integration times and pre-processing techniques. The integration time of 200 ms was recommended by Taira et al. [19], who predicted Pol in cane stalks using a portable visible-shortwave near

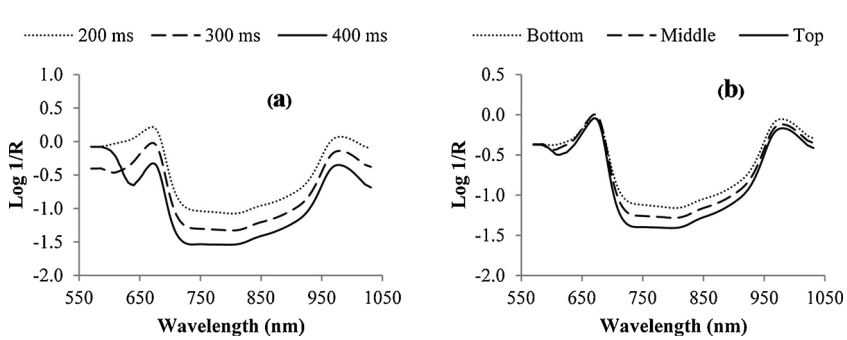

Fig. 4. Average Vis/SWNIR spectra of cane stalks with (a) different integration times and (b) different sample sections.

infrared (Vis/SWNIR) spectrometer. However, the sugarcane varieties in Thailand differ in their internal structure and percentage of sugarcane juice compared to the sugarcane varieties in Japan. The percentage of juice extraction of cane stalks in Thailand is approximately 70% [36] which is more than those in Japan. As such, the integration times of 300 and 400 ms were applied. However, longer integration times resulted in lower absorbances. The results of the PLS models for predicting fibre contents with different integration times and pre-processing techniques are shown in Table 2. Referring to the ANOVA analysis results analysed by RCBD method, the result was that the null hypothesis was rejected, meaning there was a statistically significant difference in RMSEP. The mean RMSEP differs significantly due to the effect of spectral pretreatments. The difference in RMSEP of each spectral pretreatment tested by Duncan's test is shown in Table 3. The SNV and range normalization techniques showed the highest RMSEP, and they should be avoided for model development. Meanwhile, raw spectra and spectral preprocessing of MAS gave a low RMSEP, meaning that the raw spectra and MAS are recommended because they provide the lowest RMSEP.

In this study, the model with the integration time of 300 ms, coupled with the use of raw spectra, was determined to be the best model for the prediction of fibre content. This is because it provided a low RMSEP (0.81% fibre). Furthermore, it did not need any pre-processing techniques to adjust the spectra. Although the R^2 and the r^2 of the effective model seemed to be rather low, as shown in Fig. 5, it could be used for screening and for approximate prediction [29]. This is the typical incidence for the measurement of inhomogeneous material that was affected by the sampling difference between NIR and the reference method.

Table 3
The results of ANOVA analysis for the RMSEP values of different preprocessing techniques.

Preprocessing techniques	Integration times (ms)			Average RMSEP	Standard deviation of RMSEP
	200	300	400		
Raw data	0.99	0.81	0.94	0.91a	0.092
Baseline Offset	1.02	0.96	1.09	1.02bc	0.063
Mean Centering	1.03	0.95	1.07	1.01bc	0.062
Standard Normal Variate [SNV]	1.22	1.27	1.26	1.25d	0.030
Range normalization	1.23	1.27	1.22	1.24d	0.026
MAS	0.97	0.80	0.90	0.89a	0.087
MAS, D1	1.09	0.99	1.13	1.07c	0.075
MAS, D2	1.00	0.90	0.97	0.96ab	0.052

Values with different letters within a column differ significantly ($P < 0.05$).

These results reflected the fact that a portable Vis/SWNIR spectrometer generated by PLS regression could be applied to screening the fibre content of sugarcane, which would aid breeders in their plant breeding programmes. In addition, it has the potential to be an instrument that could assess the fibre content at sugarcane transfer stations.

The results of this study, including the R^2 and the r^2, were similar to those obtained by Nawi et al. [37], who used a Vis/SWNIR spectroradiometer to predict the sugarcane fibre content, which obtained R^2 and r^2 values of 0.79 and 0.56, respectively. However, in that study, they measured the cross-section of the stalk, which is in contrast to this

Table 2
The results of the partial least squares regression of the CSS models for predicting the fibre content with different integration times and preprocessing techniques.

Integration times	Preprocessing techniques	Calibration set			Prediction set			
		F	R^2	RMSEC (%fibre)	r^2	RMSEP (%fibre)	bias$^+$ (%fibre)	RPD
200 ms	Raw data	8	0.68	0.93	0.62	0.99	0.12	1.63
	Baseline Offset	7	0.59	1.05	0.60	1.02	0.12	1.59
	Mean Centering	7	0.61	1.02	0.60	1.03	0.03	1.58
	Standard Normal Variate [SNV]	7	0.53	1.13	0.40	1.22	0.28	1.33
	Range normalization	7	0.56	1.10	0.38	1.23	0.32	1.32
	MAS	9	0.71	0.89	0.63	0.97	0.18	1.67
	MAS, D^1	9	0.63	1.00	0.54	1.09	0.13	1.49
	MAS, D^2	7	0.63	0.99	0.60	1.00	0.22	1.62
300 ms	Raw data	9	0.78	0.77	0.75	0.81	−0.03	2.00
	Baseline Offset	7	0.63	1.00	0.65	0.96	0.02	1.68
	Mean Centering	7	0.63	1.00	0.66	0.95	0.06	1.71
	Standard Normal Variate [SNV]	6	0.40	1.27	0.33	1.27	0.36	1.27
	Range normalization	6	0.41	1.26	0.36	1.27	0.27	1.28
	MAS	9	0.77	0.79	0.76	0.80	0.02	2.03
	MAS, D^1	9	0.65	0.97	0.61	0.99	0.20	1.64
	MAS, D^2	8	0.69	0.83	0.69	0.90	0.05	1.80
400 ms	Raw data	9	0.71	0.88	0.67	0.94	0.03	1.73
	Baseline Offset	7	0.61	1.02	0.55	1.09	0.06	1.49
	Mean Centering	7	0.59	1.05	0.56	1.07	0.09	1.52
	Standard Normal Variate [SNV]	7	0.41	1.27	0.35	1.26	0.33	1.28
	Range normalization	7	0.45	1.22	0.42	1.22	0.20	1.33
	MAS	9	0.72	0.87	0.68	0.90	0.11	1.79
	MAS, D^1	9	0.59	1.05	0.48	1.13	0.26	1.43
	MAS, D^2	9	0.65	0.98	0.63	0.97	0.18	1.66

F: number of PLS factors used in calibration equation.
R^2: coefficients of determination of calibration set.
r^2: coefficients of determination of prediction set.
RMSEC: the root mean square error of calibration.
RMSEP: the root mean square error of prediction.
bias: the average of difference between measured values and predicted values.
RPD: ratio of standard deviation of reference data in prediction set to RMSEP.
MAS: moving average smoothing.
D^1: first derivative.
D^2: second derivative.
$^+$by the pair *t*-test at 5% level, all the biases reported are not significantly different (Khuriyati et al. 2004).

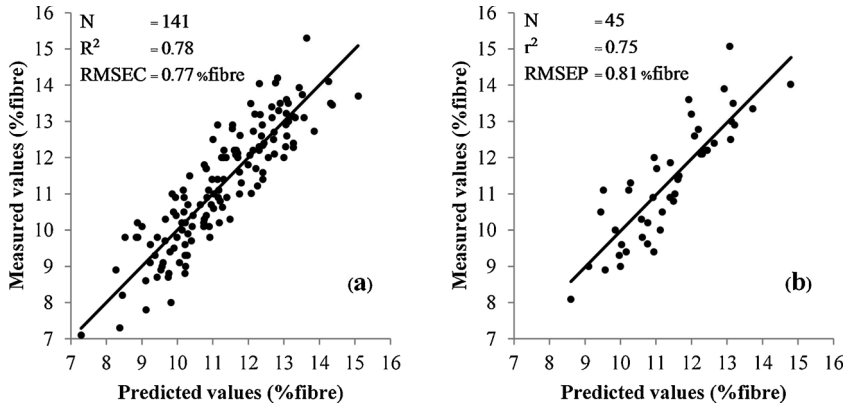

Fig. 5. Measured versus predicted values for prediction sugarcane fibre content of the CSS model using (a) calibration set, and (b) prediction set.

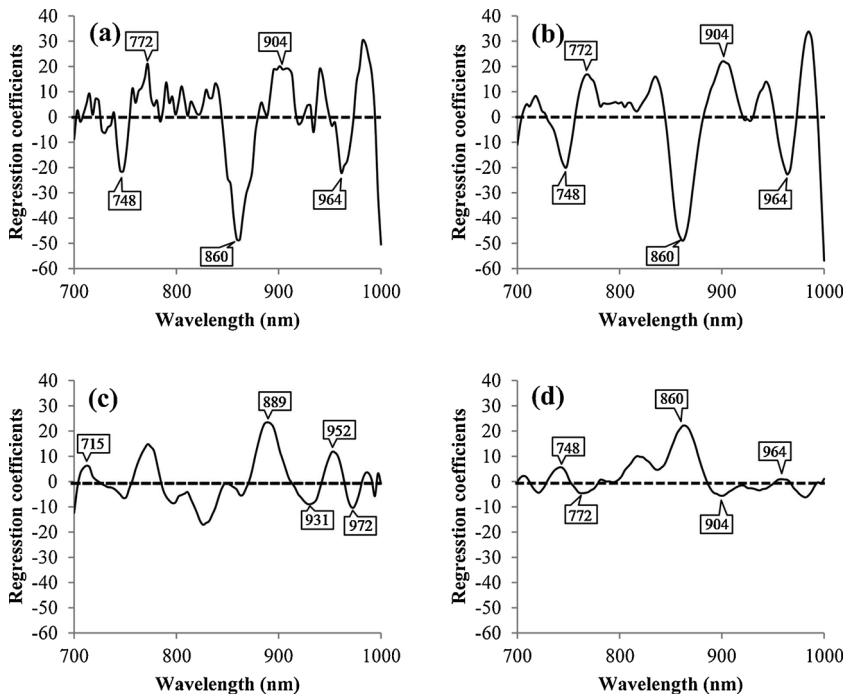

Fig. 6. Regression coefficients plot of the CSS models for the prediction of sugarcane fibre content using the integration time of 300 ms and preprocessing techniques of (a) Raw data (b) MAS (c) MAS + D^1 (d) MAS + D^2.

Table 4
Results of partial least squares regression of the ISS models developed by raw spectra for determining fibre content of different sections of sugarcane stalks.

Section of sugarcane stalks	Calibration set			Prediction set			
	F	R^2	RMSEC (%fibre)	r^2	RMSEP (%fibre)	bias (%fibre)	RPD
Top	8	0.86	0.54	0.71	0.73	−0.30	2.03
Middle	9	0.91	0.56	0.81	0.80	−0.21	2.41
Bottom	8	0.81	0.71	0.81	0.63	0.21	2.42

SD: standard deviation.
F: number of PLS factors used in calibration equation.
R^2: coefficients of determination of calibration set.
r^2: coefficients of determination of prediction set.
RMSEC: the root mean square error of calibration.
RMSEP: the root mean square error of prediction.
bias: the average of difference between measured values and predicted values.
RPD: ratio of standard deviation of reference data in prediction set to RMSEP.

study that measured the skin.

In another research by Nawi et al. [22], who predict the sugarcane fibre content by using the Vis/SWNIR spectroradiometer, the models had R^2 and r^2 of 0.92 and 0.93, respectively, for skin surface scanning. In that study, the samples had a wider range (0.2–11.5% fibre) and higher SD (2.51% fibre) of fibre content than in this research. This may have resulted from using the reference method as a secondary measurement calculated from moisture content and brix of juice values. Furthermore, their samples may have been prepared before measurement, because fibre content of 0.2% fibre has never been found in natural cane stalk. Although this research can measure the fibre content of the sugarcane stalks directly without destroying the samples, the effects of other factors that cannot be controlled in the field, such as sunlight and temperature, still require investigation. If the temperature is increased, water absorption may become stronger and may affect the prediction, such as in Kawano and Abe [38], who used the NIR instrument to predict Brix values in intact peaches and found that water absorption was easily influenced by variations in temperature. Furthermore, the samples in this study were coated with a protective layer of waxy material, of which there were various types and varied

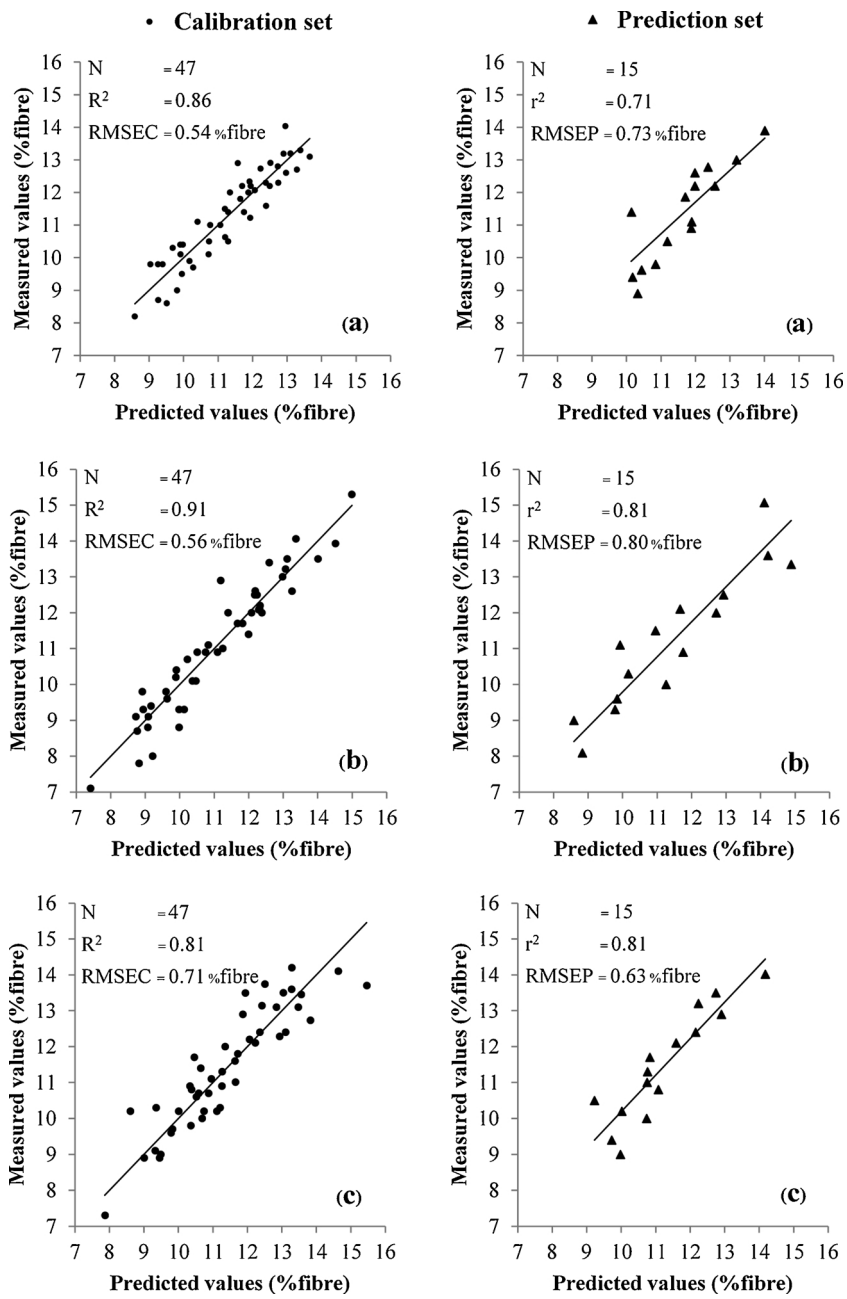

Fig. 7. Measured value versus predicted values for prediction sugarcane fibre content of the ISS models using (a) top section model, (b) middle section model and (c) bottom section model.

amounts on the samples. This waxy material may affect the spectral acquisitions and relatively low r^2. Therefore, future studies should investigate the effect of waxy material or it should be removed before measurement.

3.3. The effects of using different pre-processing techniques on the model

Fig. 6 is a plot of regression coefficients of the CSS models generated from different pre-processing techniques using an integration time of 300 ms. There was fluctuation in the plotting of the regression coefficients of the raw spectra, as shown in Fig. 6a. This incidence occurs when the spectra contain noise. The MAS revealed that it can reduce the noise in the spectra (See Fig. 6b.). However, it was also found that the model constructed using the MAS spectra provided similar results as the raw spectra. As shown in Fig. 6a and 6b, both regression coefficient plots are the same. The regression coefficient plots of the pre-processing of the MAS + D^1 (Fig. 6c) and the MAS + D^2 (Fig. 6d) show obvious peaks, making these findings similar to those Cen and He [39] and by Moghimi et al. [40]. These researchers found that the peaks became clearer after the samples were pre-treated with a derivative, even if the peak was not obvious in the original spectra. The D^1 technique made the peak obvious in the raw spectra (860 nm and 964 nm) and became zero. This is why the pre-processing of D^1 produced poor results.

Fig. 6 presents obvious peaks in the regression coefficients plot as follows: (1) at 745–760, 850–870, and 950–970 nm, which are related to water; (2) at 900–910 nm, which corresponded to carbohydrates [41]; and (3) at 765–780 nm, which are related to secondary alcohols (functional group of −CH) [42], which can be referred to the structure of lignocellulose.

3.4. PLS models established based on individual sample sections (ISS models)

In this section, each sub-section (47 samples of each of the three sub-sections) was used for separate modelling. As shown in Table 2 and 3, the model that was developed using raw spectra with an integration time of 300 ms was the best. However, the R^2 and r^2 seemed to be rather low, which might have been the effect of having different substances in the sugarcane juice in the individual sample sections. This is due to fact that the regression coefficients of the PLS model, which affect the prediction, are almost all related to water and carbohydrates or the sugar content in the sugarcane juice. According to Jintawet et al. [36], the bottom and middle sections of cane stalks mostly have a greater percentage of sugar content compared to the top section. Consequently, the total spectra using the raw spectra and integration time of 300 ms were divided into three groups based on the sample sections (i.e., the bottom, middle, and top sections). Then, models for each individual sample section were established. The results from the bottom, middle, and top sections are presented in Table 4. The middle section model provided the highest R^2 because it had the highest range and SD of fibre content. The bottom section model, which provided the lowest RMSEP of 0.63%fibre, was the best model for the individual sample section. The top section model had the lowest performance because its low sugar content provided weaker absorption information (related to the water and carbohydrates) than the bottom and middle sections. Moreover, it had the lowest range and SD of fibre contents. The scatter plot between the measured and predicted values of fibre content were validated by the top, middle, and bottom section models, respectively, as shown in Fig. 7. The ISS models for the bottom and middle sections provided a slightly higher r^2 than the CSS model, and RMSEP was lower than the CSS model. For the top section, the ISS model gave a little lower r^2 than the CSS model because the fibre content of the top section had a small range and SD compared to the fibre content of the CSS model. However, the RMSEP of the top section was still lower than that of the CSS model. These results indicated that each sub-section in the sugarcane stalk might generate the unique spectra because the composition in the sub-sections of the sugarcane stalk differ. The authors recommend the following: (1) the top section model should be used to make predictions if the top section is to be scanned, (2) the bottom section model should be used to make predictions if the bottom section is to be scanned, and (3) the middle section model should be used to make predictions if the middle section is to be scanned.

4. Conclusions

This study shows the feasibility of rapid and direct determination of the fibre content of sugarcane stalks using a portable Vis/SWNIR instrument. Among the combination of three integration times across eight pre-processing techniques, the integration time of 300 ms across raw spectra provided the best model. The ISS models provided better accuracy than the CSS model due to the different substances in sub-sections that affect the spectra. The models could be utilised for screening, which could help breeders to obtain rapid measurements and would furthermore potentially give them a tool to monitor fibre content in the field without cutting any stalks. Moreover, there is potential to measure the fibre content in the same stalk throughout the breeding programmes, which cannot be done currently. In closing, with regard to practical use in the field, future studies should focus on the influence of factors that cannot be controlled in the field, such as sunlight and temperature.

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